Software Engineering

Lab 3: Component Modelling & Architectural Pattern Selection

42. Internal Microservice Catalog & Health Portal

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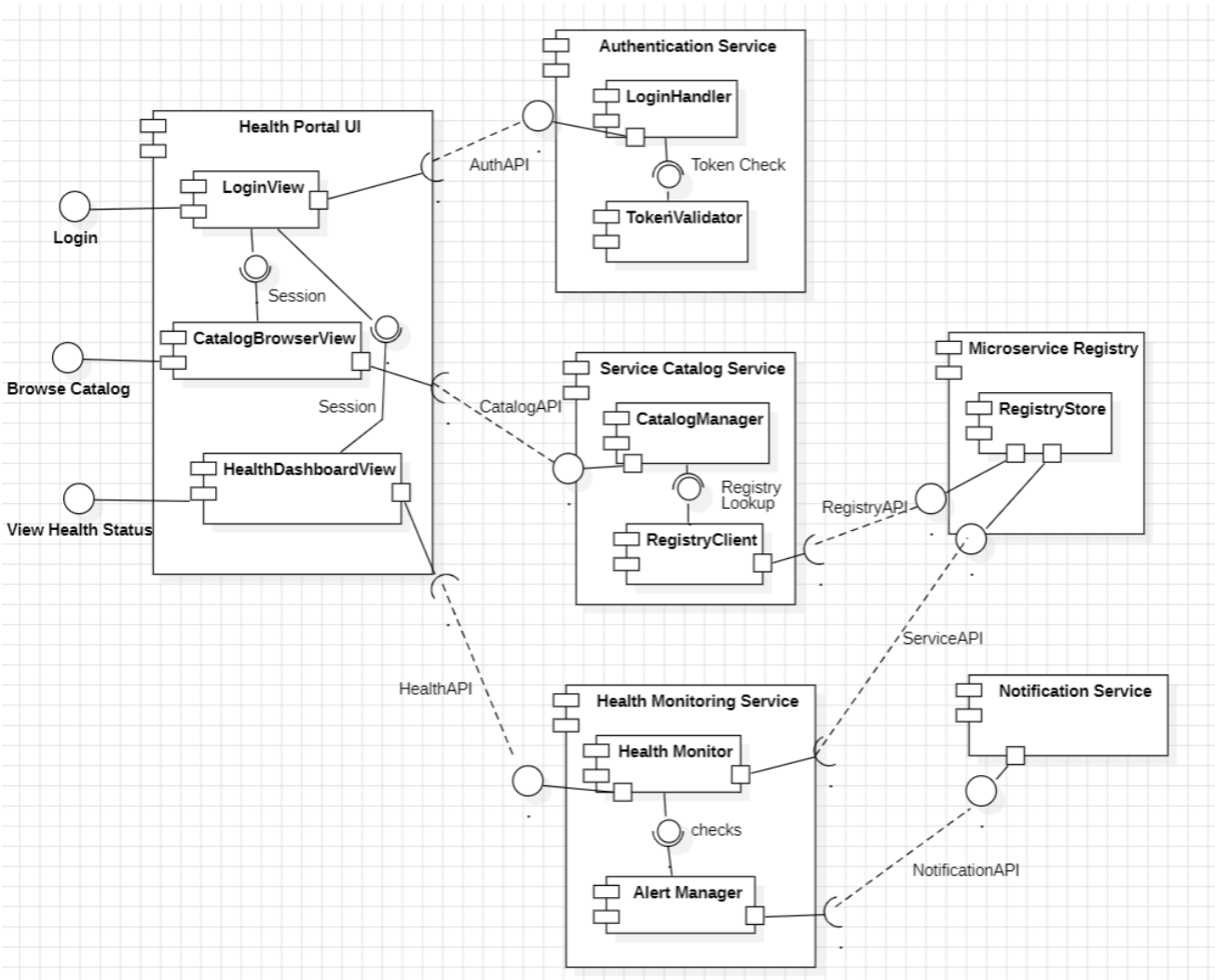
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Deliverables:

1. Component Diagram



2. Written Justification

Architecture Selection: We chose Microservices Architecture for the Internal Microservice Catalog & Health Portal System.

1. Architectural Choice:
Microservices Architecture was selected because the system consists of distinct functions that can be implemented as separate services, including Authentication, Service Catalog, Microservice Registry, Health Monitoring, and Notification. Each service communicates through well-defined APIs and can be developed, deployed, and maintained independently.

2. Reasons for the Choice:

1) Independent management: The Service Catalog, Authentication, and Health Monitoring functions can be developed and updated independently. This is useful because changes to the service catalog do not require changes to the health monitoring or authentication service.
2) Independent scalability: Health monitoring may generate frequent health-check requests for many registered microservices. With a microservices architecture, the Health Monitoring Service can be scaled independently when monitoring workload increases.

3. Security Advantage:
Security can be centralized through the Authentication Service. The Health Portal can use the Authentication API to verify users and tokens before allowing access to protected functions. Role-based access control can restrict operations such as registering or modifying microservice information to authorized users.

4. Performance Benefit:
The architecture allows frequently used or resource-intensive services to operate independently. For example, Health Monitoring can perform health checks and process service-status information without blocking catalog or authentication operations. Individual services can also be scaled according to their workload, improving responsiveness as the number of monitored services grows.